Harshita Rathore

Python Backend Developer | FastAPI | Generative AI & RAG

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Professional Summary

Entry-level Python Backend Developer and B.Tech Computer Science graduate with hands-on project experience building API-driven applications using FastAPI, SQL, and PostgreSQL. Completed internships in Generative AI and Machine Learning, with projects integrating Large Language Model (LLM) APIs, Retrieval-Augmented Generation (RAG), and agentic workflows. Familiar with vector search using FAISS, database operations with SQLAlchemy, JWT-based authentication, and containerization fundamentals.

Technical Skills

Programming & Backend: Python, FastAPI, REST APIs, JWT Authentication

GenAI & Applied AI: Generative AI, Retrieval-Augmented Generation (RAG) Pipelines, Large Language Model (LLM) Integration, Agentic AI (CrewAI), Google Gemini API, Groq, Basic Machine Learning, Basic Natural Language Processing (NLP)

Data & Databases: SQL, PostgreSQL, MySQL (RDBMS), SQLAlchemy (ORM), FAISS (Vector Database)

Cloud & DevOps: Docker (Basic), GitHub Actions, AWS EC2 (Basic), CI/CD Fundamentals

Tools: Git, GitHub, VS Code, Google Colab, Postman, Swagger UI

Professional Experience

Datagami (NASSCOM x IBM Initiative) *Jan 2026 – Apr 2026*

Generative AI & Agentic Systems Intern

- Developed a Retrieval-Augmented Generation (RAG) chatbot using FastAPI, a FAISS vector store, and Google Gemini to answer contextual questions about a retail product catalog.
- Built a four-agent CrewAI workflow (Planner, Researcher, Analyst, and Critic agents) using Groq LLM and DuckDuckGo search for query decomposition and synthesis.
- Reduced Docker image size by approximately 60% by replacing Hugging Face embeddings with Google Generative AI Embeddings, lowering startup overhead.
- Containerized with Docker; automated image builds and deployment via a GitHub Actions CI/CD pipeline.

Medi-Caps University (In-House Research Lab) *Jun 2025 – Jul 2025*

AI / ML Intern

- Gained introductory exposure to Python-based ML workflows: data preparation, model training, and basic evaluation.
- Practiced foundational supervised learning concepts and explored image-classification workflows in guided tasks.

Projects

Retail RAG Chatbot + Agent Mode GitHub

Tech Stack: FastAPI, FAISS, Gemini, CrewAI, Groq, DuckDuckGo, Docker, GitHub Actions, AWS EC2

- Designed an end-to-end RAG pipeline with a FAISS vector store for semantic retrieval; added a modular Agent Mode toggle integrating CrewAI multi-agent workflows.
- Containerized with Docker and deployed on AWS EC2; automated build and push using GitHub Actions.

StockSphere: Intelligent Inventory Management with Generative AI GitHub

Tech Stack: Python, FastAPI, React, PostgreSQL, SQLAlchemy, JWT, Google Gemini API

- Developed a full-stack inventory app with product CRUD operations using React, FastAPI, and PostgreSQL.
- Used SQLAlchemy for database operations; implemented JWT authentication with user-specific inventory retrieval.
- Integrated the Gemini API for AI-assisted product description generation and implemented Python-based stock classification with an inventory-advice endpoint.

Education

Medi-Caps University, Indore *2022 – 2026*

Bachelor of Technology (B.Tech) in Computer Science Engineering *CGPA: 8.26*

Bal Niketan Higher Secondary School, Indore *2020 – 2022*

Class XII, CBSE *Class X: 94.67% | Class XII: 73.6%*

Certifications

- SQL – HackerRank
- Python Essentials – Cisco Networking Academy
- Generative AI – AWS
- Introduction to Large Language Models – Google Cloud Skills Boost